

Microsoft Excel

Complete Study Notes

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Sameeksha	2026

Basics • Formulas • Functions • Charts • PivotTables • Data Tools • Shortcuts

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1. Introduction to Microsoft Excel

Microsoft Excel is a powerful spreadsheet application developed by Microsoft, part of the Microsoft Office/365 suite. It is used worldwide for data analysis, financial modelling, reporting, automation, and much more.

Key Uses of Excel

- Financial analysis – budgets, forecasts, P&L; statements
- Data organisation – storing and managing large datasets
- Statistical analysis – mean, median, standard deviation, regression
- Visualisation – charts, graphs, sparklines
- Automation – Macros, VBA, Power Query
- Business reporting – dashboards, KPI trackers

2. Excel Interface & Navigation

When you open Excel, the interface consists of several key components:

Component	Description
Ribbon	Toolbar with tabs: Home, Insert, Page Layout, Formulas, Data, Review, View
Quick Access Toolbar	Customisable toolbar for frequently used commands (top-left)
Name Box	Displays the address of the currently selected cell (e.g., A1)
Formula Bar	Shows and allows editing of the content/formula in the active cell
Worksheet Area	The grid made of rows (numbered) and columns (lettered)
Sheet Tabs	Navigate between multiple worksheets at the bottom
Status Bar	Shows Sum, Average, Count of selected cells at the bottom
Zoom Slider	Adjust view zoom level (bottom-right corner)

Navigation Shortcuts

Keys	Action
Arrow Keys	Move one cell in that direction
Ctrl + Arrow	Jump to the last non-empty cell in a direction
Ctrl + Home	Go to cell A1
Ctrl + End	Go to the last used cell in the sheet

Keys	Action
Ctrl + G / F5	Go To dialog – navigate to a specific cell
Page Up / Down	Scroll one screen up or down

3. Workbooks, Worksheets & Cells

Workbook: An Excel file (.xlsx, .xlsm, .xls). One workbook can contain many worksheets.

Worksheet (Sheet): A single spreadsheet tab within a workbook.

Cell: The intersection of a row and a column. Each cell has a unique address (e.g., B5 = Column B, Row 5).

Range: A group of cells (e.g., A1:D10 covers columns A–D, rows 1–10).

Cell Reference Types

Reference	Example	Behaviour when copied
Relative	=A1	Row and column both adjust
Absolute	=\$A\$1	Row and column stay fixed
Mixed (row fixed)	=A\$1	Row stays fixed, column adjusts
Mixed (col fixed)	=\$A1	Column stays fixed, row adjusts

Tip: Press F4 while editing a cell reference to cycle through all reference types.

4. Data Entry & Formatting

Types of Data

- **Numbers** – integers, decimals, percentages, currency
- **Text** – labels, names, descriptions (left-aligned by default)
- **Dates & Times** – stored as serial numbers internally
- **Logical** – TRUE / FALSE
- **Formulas** – always start with '='

Formatting Options (Home Tab)

Format	Purpose
Number Format	Currency, Percentage, Date, Time, Scientific, Custom

Format	Purpose
Font	Typeface, Size, Bold, Italic, Underline, Colour
Alignment	Left, Centre, Right, Merge & Centre, Wrap Text, Indent
Borders	Cell borders (outline, inside, colour, style)
Fill Colour	Highlight cells with background colour
Format Cells (Ctrl+1)	Full dialog box for all formatting options

Useful Paste Special Options (Ctrl + Alt + V)

- Values Only – paste result without formula
- Formats Only – paste only styling
- Transpose – swap rows and columns
- Operations – Add/Subtract/Multiply/Divide onto existing values

5. Formulas & Operators

A formula begins with '=' and can include cell references, values, operators, and functions.

Arithmetic Operators

Operator	Operation	Example	Result
+	Addition	=5+3	8
-	Subtraction	=10-4	6
*	Multiplication	=6*7	42
/	Division	=20/4	5
^	Exponentiation	=2^10	1024
%	Percentage	=50%	0.5
&	Text concatenation	="Hello"&" "&"World"	Hello World

Comparison Operators

Operator	Meaning	Example
=	Equal to	=A1=B1
<>	Not equal to	=A1<>B1
>	Greater than	=A1>100

Operator	Meaning	Example
<	Less than	=A1<0
>=	Greater than or equal	=A1>=50
<=	Less than or equal	=A1<=50

6. Essential Excel Functions

Math & Statistical Functions

Function	Syntax	Description
SUM	=SUM(A1:A10)	Adds all values in range
AVERAGE	=AVERAGE(A1:A10)	Returns arithmetic mean
COUNT	=COUNT(A1:A10)	Counts numeric cells
COUNTA	=COUNTA(A1:A10)	Counts non-empty cells
COUNTBLANK	=COUNTBLANK(A1:A10)	Counts empty cells
MAX	=MAX(A1:A10)	Returns largest value
MIN	=MIN(A1:A10)	Returns smallest value
MEDIAN	=MEDIAN(A1:A10)	Returns middle value
MODE	=MODE(A1:A10)	Returns most frequent value
STDEV	=STDEV(A1:A10)	Sample standard deviation
ROUND	=ROUND(A1,2)	Rounds to n decimal places
ROUNDUP	=ROUNDUP(A1,0)	Always rounds up
ROUNDDOWN	=ROUNDDOWN(A1,0)	Always rounds down
ABS	=ABS(A1)	Absolute value
MOD	=MOD(A1,3)	Remainder after division
POWER	=POWER(2,8)	Raises number to a power
SQRT	=SQRT(A1)	Square root
SUMIF	=SUMIF(A1:A10,">100")	Sum cells meeting a condition
SUMIFS	=SUMIFS(C1:C10,A1:A10,"X",B1:B10,">5")	Sum with multiple criteria
COUNTIF	=COUNTIF(A1:A10,"Apple")	Count cells meeting a condition

Function	Syntax	Description
COUNTIFS	=COUNTIFS(A1:A10,"X",B1:B10,">0")	Count with multiple criteria
AVERAGEIF	=AVERAGEIF(A1:A10,">50")	Average cells meeting a condition
RANK	=RANK(A1,A1:A10,0)	Rank of a value in a list
LARGE	=LARGE(A1:A10,2)	Returns n-th largest value
SMALL	=SMALL(A1:A10,2)	Returns n-th smallest value

7. Logical & Lookup Functions

Logical Functions

Function	Syntax	Description
IF	=IF(condition, val_true, val_false)	Returns one of two values based on condition
IFS	=IFS(cond1,val1,cond2,val2,...)	Multiple conditions (Excel 2019+)
AND	=AND(A1>0, B1<100)	TRUE if all conditions are true
OR	=OR(A1="Yes", B1="Y")	TRUE if any condition is true
NOT	=NOT(A1=0)	Reverses TRUE/FALSE
IFERROR	=IFERROR(formula, "Error")	Returns alternate value on error
IFNA	=IFNA(VLOOKUP(...), "Not Found")	Handles #N/A errors specifically
SWITCH	=SWITCH(A1,1,"One",2,"Two")	Matches value to list (Excel 2019+)

Lookup & Reference Functions

Function	Syntax	Description
VLOOKUP	=VLOOKUP(val, table, col, 0)	Vertical lookup – searches first column
HLOOKUP	=HLOOKUP(val, table, row, 0)	Horizontal lookup – searches first row
INDEX	=INDEX(range, row, col)	Returns value at a specific position
MATCH	=MATCH(val, range, 0)	Returns relative position of a value
INDEX+MATCH	=INDEX(B1:B10,MATCH(val,A1:A10,0))	Flexible alternative to VLOOKUP
XLOOKUP	=XLOOKUP(val,lookup_arr,return_arr)	Modern lookup (Excel 365/2021)
CHOOSE	=CHOOSE(2,"A","B","C")	Returns value from a list by index

Function	Syntax	Description
OFFSET	=OFFSET(A1,2,3,1,1)	Returns reference offset from a start cell
INDIRECT	=INDIRECT("A"&B1)	Returns reference from a text string
ROW / COLUMN	=ROW(A5) → 5	Returns row or column number of a cell
ROWS / COLUMNS	=ROWS(A1:A10) → 10	Returns count of rows or columns in range

8. Date & Text Functions

Date & Time Functions

Function	Example	Description
TODAY()	=TODAY()	Returns today's date
NOW()	=NOW()	Returns current date and time
DATE	=DATE(2026,4,21)	Creates a date from year, month, day
DAY/MONTH/YEAR	=YEAR(A1)	Extracts day, month, or year
WEEKDAY	=WEEKDAY(A1,2)	Returns day of week (1=Monday if type=2)
NETWORKDAYS	=NETWORKDAYS(A1,B1)	Counts working days between two dates
EDATE	=EDATE(A1,3)	Adds/subtracts months from a date
EOMONTH	=EOMONTH(A1,0)	Last day of month
DATEDIF	=DATEDIF(A1,B1,"Y")	Difference in years/months/days

Text Functions

Function	Example	Description
LEN	=LEN(A1)	Number of characters in text
LEFT	=LEFT(A1,5)	First n characters
RIGHT	=RIGHT(A1,3)	Last n characters
MID	=MID(A1,3,4)	Characters from position 3, length 4
FIND	=FIND("@",A1)	Position of a substring (case-sensitive)
SEARCH	=SEARCH("hello",A1)	Position of substring (case-insensitive)
SUBSTITUTE	=SUBSTITUTE(A1,"old","new")	Replaces occurrences of text

Function	Example	Description
REPLACE	=REPLACE(A1,2,3,"XY")	Replaces by position
TRIM	=TRIM(A1)	Removes extra spaces
UPPER/LOWER/PROPER	=PROPER(A1)	Changes text case
CONCATENATE / &	=A1&" "&B1	Joins text strings
TEXTJOIN	=TEXTJOIN(", ",TRUE,A1:A5)	Joins range with delimiter (2019+)
TEXT	=TEXT(A1,"DD-MMM-YYYY")	Formats a number as text
VALUE	=VALUE("1234")	Converts text to number

9. Charts & Graphs

Charts visually represent data. Select your data range, then go to **Insert** → **Charts**.

Common Chart Types

Chart Type	Best Used For
Column / Bar Chart	Comparing categories side by side
Line Chart	Showing trends over time
Pie / Doughnut Chart	Showing parts of a whole (proportions)
Area Chart	Cumulative values over time
Scatter Plot	Correlation between two numeric variables
Bubble Chart	Scatter with a third variable (bubble size)
Histogram	Distribution of numerical data
Box & Whisker	Statistical spread and outliers
Waterfall Chart	Running total with positive/negative values
Combo Chart	Mix of two chart types (e.g., Line + Column)
Sparklines	Tiny charts inside cells for quick trend view

Chart Formatting Tips

- Double-click any chart element to format it
- Use Chart Title, Axis Titles, Data Labels, Legend via the '+' button
- Change chart type via right-click → Change Chart Type
- Move chart to a new sheet: Design → Move Chart

- Apply chart styles from the Design tab for quick professional looks

10. PivotTables & PivotCharts

A PivotTable summarises large datasets dynamically without writing formulas. Insert → PivotTable.

PivotTable Areas

Area	Purpose
Rows	Field values displayed as row labels
Columns	Field values displayed as column headers
Values	Numeric data to aggregate (SUM, COUNT, AVERAGE, MAX, MIN)
Filters	Filters the entire PivotTable by the chosen field

PivotTable Tips

- Refresh data: Right-click → Refresh (or Alt + F5)
- Group dates by Month/Quarter/Year by right-clicking date field
- Show values as % of Total, Running Total, Rank etc. (Value Field Settings)
- Slicers: Insert → Slicer – visual filter buttons for PivotTables
- Timelines: Insert → Timeline – filter by date ranges interactively
- PivotCharts are linked charts based on PivotTable data

11. Data Validation & Protection

Data Validation (Data tab → Data Validation) restricts what users can enter in a cell.

Validation Types

Type	Description / Example
Whole Number	Only integers within a range (e.g., 1 to 100)
Decimal	Numbers with decimals within bounds
List	Dropdown from a range or comma-separated values
Date / Time	Only dates within a specified range
Text Length	Limit characters (e.g., max 50 characters)
Custom	Any formula that returns TRUE/FALSE

Sheet & Workbook Protection

- Protect Sheet (Review → Protect Sheet) – lock cells from editing
- Unlock specific cells before protecting: Format Cells → Protection → uncheck Locked
- Protect Workbook – prevents adding/deleting/moving sheets
- Encrypt with Password: File → Info → Protect Workbook → Encrypt with Password
- Mark as Final – makes workbook read-only (soft protection)

12. Sorting, Filtering & Tables

Sorting

- Quick Sort: Data → Sort A to Z or Z to A
- Multi-level Sort: Data → Sort → Add Level (sort by multiple columns)
- Sort by Cell Colour or Font Colour (useful with conditional formatting)
- Custom List Sort: sort by months, days, or any custom order

AutoFilter

- Enable: Data → Filter (Ctrl + Shift + L)
- Filter by value, colour, text/number/date conditions
- Filter by search box for quick keyword lookup
- Advanced Filter: filter to another location using criteria range

Excel Tables (Ctrl + T)

- Auto-expands when new rows/columns are added
- Structured references: =Table1[Sales] instead of column letters
- Built-in Total Row with aggregation options
- Table Styles for instant professional formatting
- Automatic filter buttons on headers

13. Conditional Formatting

Conditional Formatting automatically formats cells based on their values or formulas. Found on the Home tab.

Option	Description
Highlight Cell Rules	Highlight cells > < = value, duplicates, between range
Top / Bottom Rules	Highlight top 10, bottom 10%, above average, etc.
Data Bars	In-cell bar chart showing relative value

Option	Description
Color Scales	Gradient colour based on value (2- or 3-colour scale)
Icon Sets	Traffic lights, arrows, stars based on thresholds
New Rule (Formula)	Custom rule using any formula returning TRUE/FALSE
Manage Rules	Edit, delete, reorder, or copy existing rules

Example formula rule: =A1>AVERAGE(\$A\$1:\$A\$100) – highlights cells above average.

14. What-If Analysis & Data Tools

What-If Analysis (Data tab)

Tool	Description
Goal Seek	Finds the input value needed to achieve a specific formula result
Scenario Manager	Save and compare multiple sets of input values
Data Table	One-variable or two-variable sensitivity analysis table
Solver	Complex optimisation with constraints (Add-in required)

Other Data Tools

Tool	Description
Remove Duplicates	Delete duplicate rows from a range or table
Text to Columns	Split cell content by delimiter or fixed width
Flash Fill	Auto-fills pattern (Ctrl + E) – great for reformatting data
Consolidate	Combine data from multiple ranges
Power Query	Advanced data import, transform, and load (ETL)
Power Pivot	In-memory data model for large datasets and DAX formulas

15. Macros & VBA Basics

Macros automate repetitive tasks. VBA (Visual Basic for Applications) is the programming language behind Excel macros.

Recording a Macro

- View → Macros → Record Macro (or click the icon in the status bar)
- Give the macro a name and optional keyboard shortcut
- Perform the steps you want to automate
- Stop recording: View → Macros → Stop Recording
- Run: View → Macros → View Macros → Run

Basic VBA Structure

```
Sub MacroName() - declares a subroutine
' This is a comment
Range("A1").Value = "Hello"
MsgBox "Task Complete!"
End Sub
```

Common VBA Objects

Object	Example	Description
Workbook	ThisWorkbook.Save	The Excel file
Worksheet	Sheets("Sheet1").Select	A single sheet
Range	Range("A1:B5").ClearContents	Cell or range of cells
Cells	Cells(1,1).Value = 100	Cell by row, column number
MsgBox	MsgBox "Done!"	Displays a popup message
InputBox	x = InputBox("Enter value:")	Gets user input
For Each Loop	For Each cell In Range("A1:A10")	Iterates over each cell
If Statement	If Range("A1").Value > 0 Then	Conditional logic

16. Keyboard Shortcuts

Shortcut	Action	Shortcut	Action
Ctrl + N	New workbook	Ctrl + S	Save
Ctrl + O	Open workbook	Ctrl + W	Close workbook
Ctrl + Z / Y	Undo / Redo	Ctrl + P	Print
Ctrl + C / X / V	Copy / Cut / Paste	Ctrl + D	Fill Down
Ctrl + R	Fill Right	Ctrl + F	Find
Ctrl + H	Find & Replace	Ctrl + G	Go To
Ctrl + 1	Format Cells dialog	Ctrl + T	Create Table

Ctrl + Shift + L	Toggle AutoFilter	Ctrl + ;	Insert today's date
Ctrl + Shift + :	Insert current time	Ctrl + `	Show/hide formulas
Alt + =	AutoSum	F2	Edit active cell
F4	Repeat last action / toggle ref	F5	Go To dialog
F11	Create chart on new sheet	F12	Save As
Ctrl + Shift + +	Insert row/column	Ctrl + -	Delete row/column
Ctrl + Space	Select entire column	Shift + Space	Select entire row
Ctrl + Shift + End	Extend to last used cell	Ctrl + A	Select all

17. Tips & Best Practices

- Always freeze the top row (View → Freeze Panes → Freeze Top Row) for long datasets.
- Use named ranges (Formulas → Name Manager) for more readable formulas.
- Avoid storing data in merged cells – they break sorting and filtering.
- Use Excel Tables (Ctrl+T) for automatic range expansion and structured references.
- Add a data entry sheet separate from your analysis/reporting sheets.
- Protect formula cells while leaving input cells editable.
- Use IFERROR to handle errors gracefully in formulas shown to users.
- Audit formulas using Formulas → Trace Precedents / Trace Dependents.
- Use Print Area (Page Layout → Print Area) to define exactly what prints.
- Save files as .xlsx (no macros) or .xlsm (with macros).
- Use Ctrl+` to toggle formula view for debugging.
- Document your workbook with a 'README' sheet for other users.
- Use Power Query for repeatable data import and transformation.
- Back up your work regularly and use version control for important files.
- Prefer INDEX+MATCH or XLOOKUP over VLOOKUP – they are more flexible and faster.

End of Notes – Sameeksha | 2026